

Dwijesh Dookraz

SOFTWARE ENGINEER · BACKEND · AI SYSTEMS · APPLIED MACHINE LEARNING

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EDUCATION

University of Southampton

2021 – 2024

BSc Computer Science · First Class Honours

Dissertation: **Machine Learning-Based Heart Rate Measurement Using rPPG — 82%**

Machine Learning Technologies **80%** Social Computing Techniques **83%** Cloud Application Development **79%**

Software Engineering Group Project **73%**

EXPERIENCE

Nusmark

2024 – Present

Backend Engineer

- Built bidirectional Google Calendar and Outlook sync using Python, OAuth2, and FastAPI — handling webhook subscriptions, sync token refresh, and transactional Firestore writes to eliminate duplicate calendar events in production
- Developed WhatsApp AI assistant using Python, Flask, and WhatsApp Cloud API with NLP-driven task extraction across text, audio, and image pipelines — deployed on GCP Cloud Run for automated event creation and user onboarding
- Engineered multi-channel notification service using Python, Firestore, and Google Calendar API integrated with Microsoft Graph API, routing reminders and mobile push across cross-provider event lifecycle

PROJECTS

rPPG Heart Rate Estimation · Final-Year Dissertation

2025

- Implemented DeepPhys dual-stream attention CNN using PyTorch and OpenCV to estimate heart rate from facial video without sensors — achieving Pearson $r = 0.87$ and MAE 8.85 BPM on the UBFC-rPPG benchmark
- Collected 15-subject dataset spanning Fitzpatrick skin types II–VI, combined with UBFC-rPPG in Python and NumPy training pipelines to address demographic bias — recovering cross-population correlation from $r = -0.23$ to $r = +0.59$
- Applied Optuna hyperparameter search with subject-aware k-fold cross-validation in PyTorch, benchmarking against ICA, PCA, frequency-domain, and band-pass signal-processing baselines

Hybrid Recommender System

2026

- Built matrix factorisation and hybrid collaborative filtering (user-CF + item-CF + content signals) from scratch using Python and NumPy with streaming SGD, Nesterov momentum, and Optuna Bayesian search — achieving MAE 0.587 on 20M MovieLens ratings without full dataset in RAM

VS Code Portfolio · dwijesh.dev

2024 – 2026

- Built browser-based VS Code shell using Next.js 15, TypeScript, and React with Monaco Editor — integrating GitHub API via server-side route handlers, Sentry error monitoring, Vercel Analytics, and Playwright UI test suite for production deployment
- Designed proxy API route in Next.js to server-side fetch shields.io and skillicons.dev badges, bypassing browser Content Security Policy restrictions for dynamic badge rendering in the editor preview

Gene Expression Analysis — Osteosarcoma

2025

- Built ML pipeline on GSE1000 microarray data using Python and scikit-learn — ANOVA feature selection across 22,283 probes, Shrinkage LDA with LOOCV (balanced accuracy 0.80), Random Forest (2,000 trees), and k-means clustering ($k=3$) to confirm 26 ECM genes downregulated at 32h via GO enrichment